Skills Summary

Adding, Subtracting, and Multiplying Polynomials

Use this reference while completing your worksheet or when studying.

Adding and subtracting are about *matching* terms; multiplying is about *pairing* them. Decide which of the two you are doing before you write anything, because the rules for exponents are opposite: like terms keep their exponent, multiplied terms add theirs.

1. Adding polynomials

Rule: drop the parentheses and combine like terms.

Like terms have *identical* variable parts — same letters, same exponents. $4x^2$ and $9x^2$ are like; $4x^2$ and $4x$ are not; x^2y and xy^2 are not. Only the coefficients are added; the variable part never changes.

Example: $(2x^2 + 3x - 1) + (4x^2 - 5x + 6)$
Step 1. Addition changes no signs, so this is purely a sorting job — group the terms by degree.
Step 2. $2x^2 + 4x^2 = 6x^2$; $3x - 5x = -2x$; $-1 + 6 = 5$.
$\boldsymbol{6x^2 - 2x + 5}$

Try it: $(x^2 - 7x + 2) + (3x^2 + 7x - 9)$

check: $\boldsymbol{4x^2 - 7}$

2. Subtracting polynomials

Rule: $A - B = A + (-B)$. Distribute the minus sign to *every* term of B, then add as usual.

$$-(ax^2 + bx + c) = -ax^2 - bx - c$$

Writing the rewritten sum on its own line is what prevents the classic one-term-only sign error.

Example: $(5x^2 - 2x + 8) - (3x^2 + 4x - 1)$
Step 1. A subtraction is an addition in disguise, so negate the whole second polynomial before combining anything.
Step 2. $(5x^2 - 2x + 8) + (-3x^2 - 4x + 1)$, so $2x^2$, $-6x$, 9.
$\boldsymbol{2x^2 - 6x + 9}$

Try it: $(4x^2 + x - 3) - (x^2 - 5x + 7)$

check: $\boldsymbol{3x^2 + 6x - 10}$

3. Multiplying by a monomial

Rule: distribute the monomial across every term, multiplying coefficients and *adding* exponents on each variable separately.

$$x^m \cdot x^n = x^{m+n}$$

A negative monomial flips the sign of every term it touches — all of them, not just the first.

Example: $3x^2(2x-5)$
Step 1. One factor is a single term, so the distributive property applies directly — no FOIL needed.
Step 2. $3x^2 \cdot 2x = 6x^3$ (exponents $2+1=3$); $3x^2 \cdot (-5) = -15x^2$.
$\boldsymbol{6x^3 - 15x^2}$

Try it: $-2x(4x^2-3x+1)$

check: $\boldsymbol{-8x^3 + 6x^2 - 2x}$

4. Multiplying two binomials

Rule: every term of the first factor times every term of the second — four products, then combine the two middle ones.

$$(a+b)(c+d) = ac + ad + bc + bd$$

(!) $(x+k)^2$ is $(x+k)(x+k)$, which is $x^2+2kx+k^2$ — never x^2+k^2. The middle term is the half that gets dropped.

Example: $(x+3)(x-7)$
Step 1. Both factors have two terms, so there are four products to write before anything can be combined.
Step 2. $x^2 - 7x + 3x - 21$, and the middle terms give $-7x + 3x = -4x$.
$\boldsymbol{x^2 - 4x - 21}$

Try it: $(2x-1)(x+5)$

check: $\boldsymbol{2x^2 + 9x - 5}$